Lecture 6b: Reductions Recap

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Lecture 6 Recap

Reductions

Informally

Reducibility Use knowledge about complexity of one language to reason about the complexity of other in an easy way

Reduction Way to show that to solve one problem (A), it is sufficient to solve another problem (B)

A reduces to B...

- Many examples (e.g. calculate area of rectangle reduces to calculate its width and height)
- Reductions quantify relative hardness of problems
  - If problem B is easy then problem A is easy too
  - If problem A is hard then problem B is hard too

A reduction f from triangle to rectangle area

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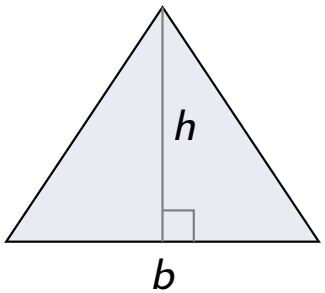
Triangle-Area $=$

$\{\langle h, b, t\rangle \in \{0,1\}^* \mid h * b/2 \geq t\}$



In language if area is at least t

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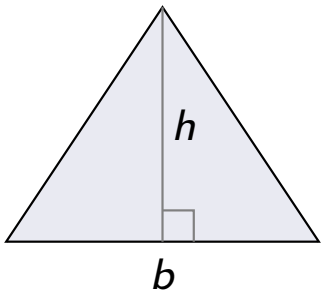

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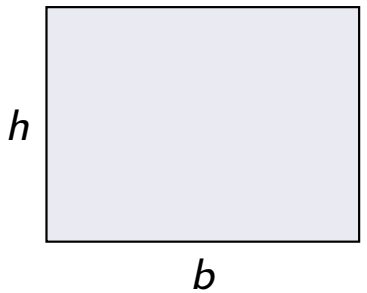

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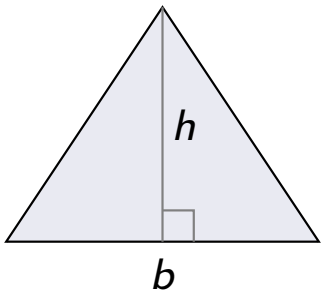

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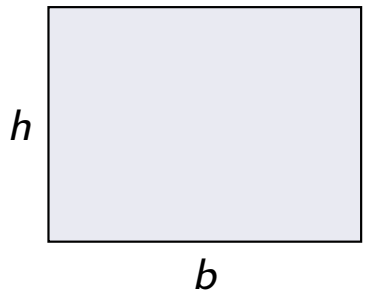

In language if area is at least r

Suppose we have a Turing machine TM_{RA} for deciding Rectangle-Area. How can we use it form a decider TM_{TA} for Triangle-Area?

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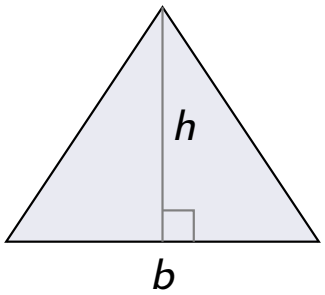

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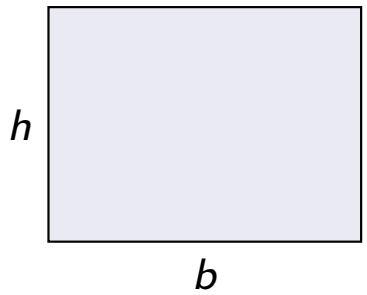

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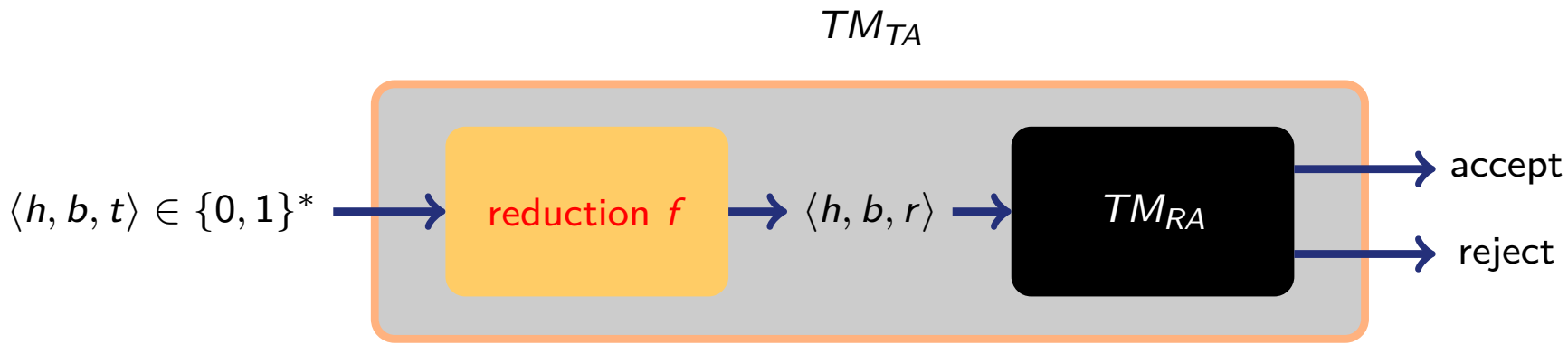

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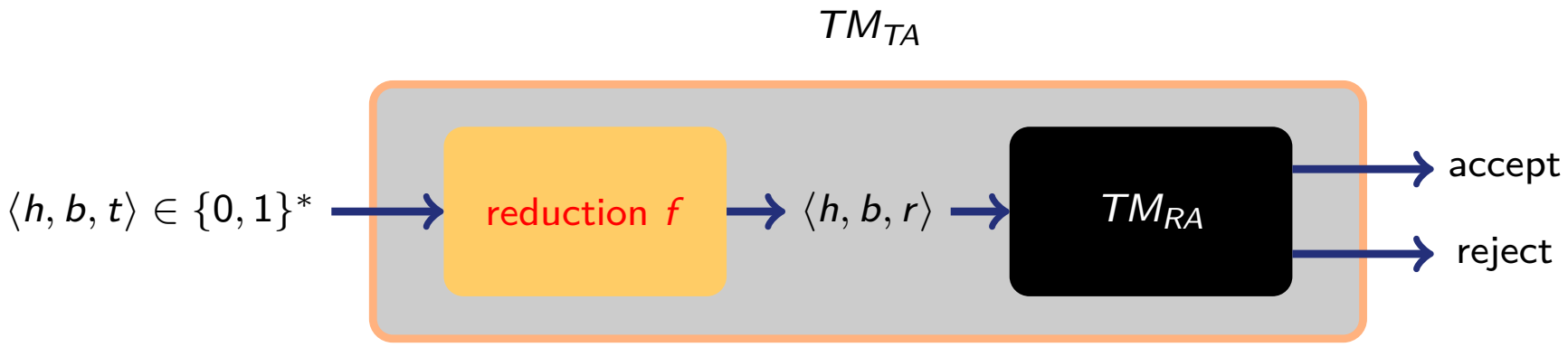

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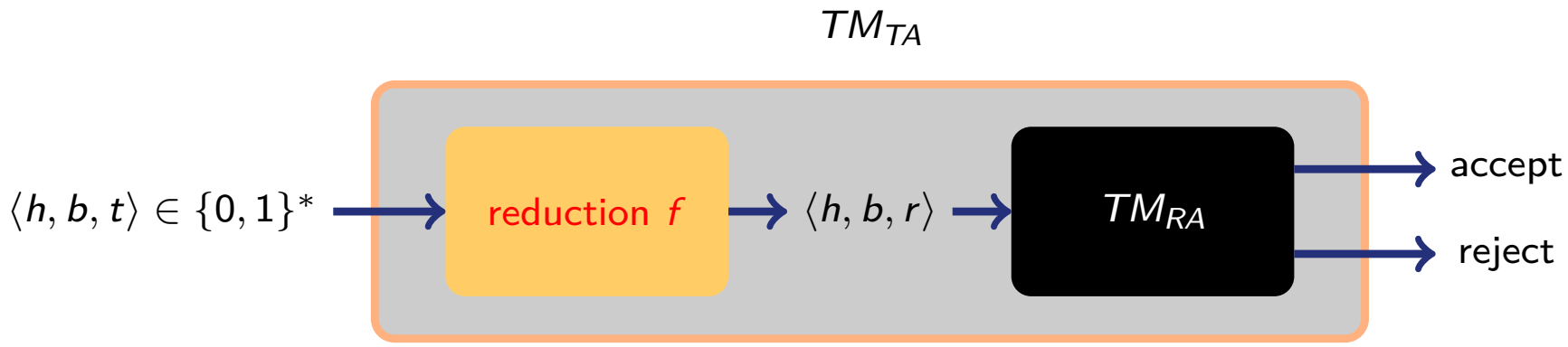

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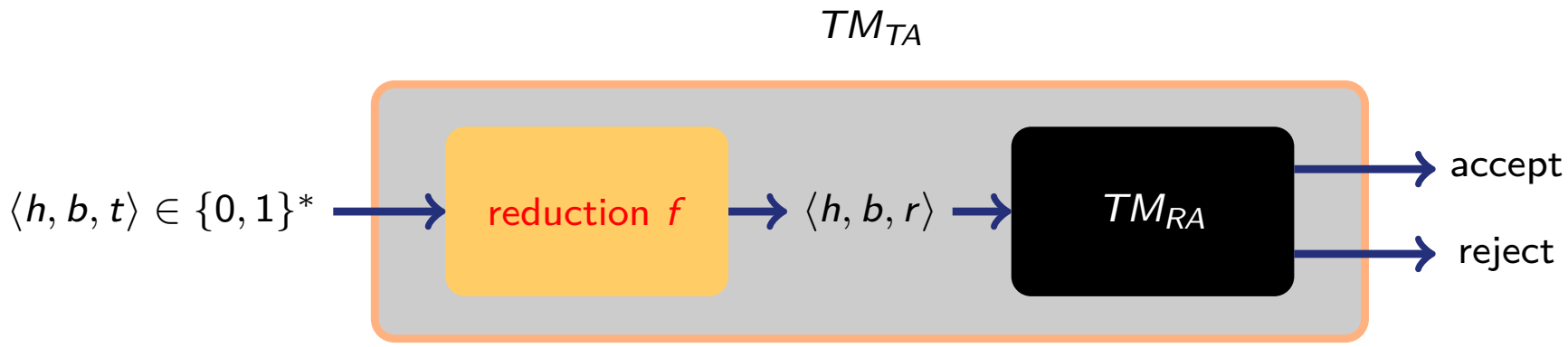

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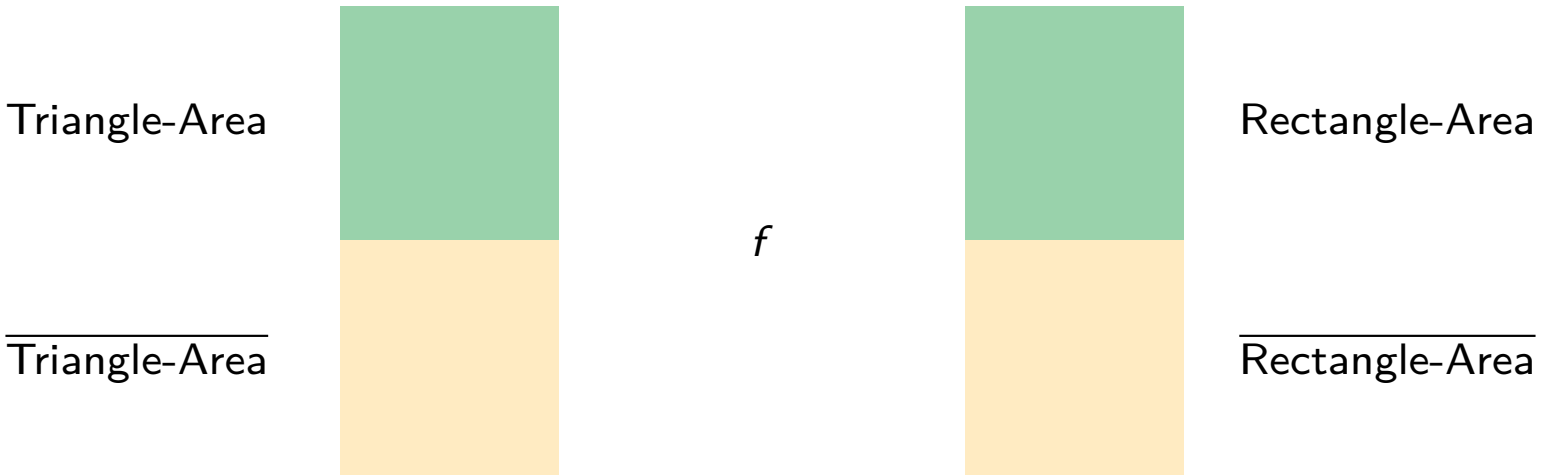

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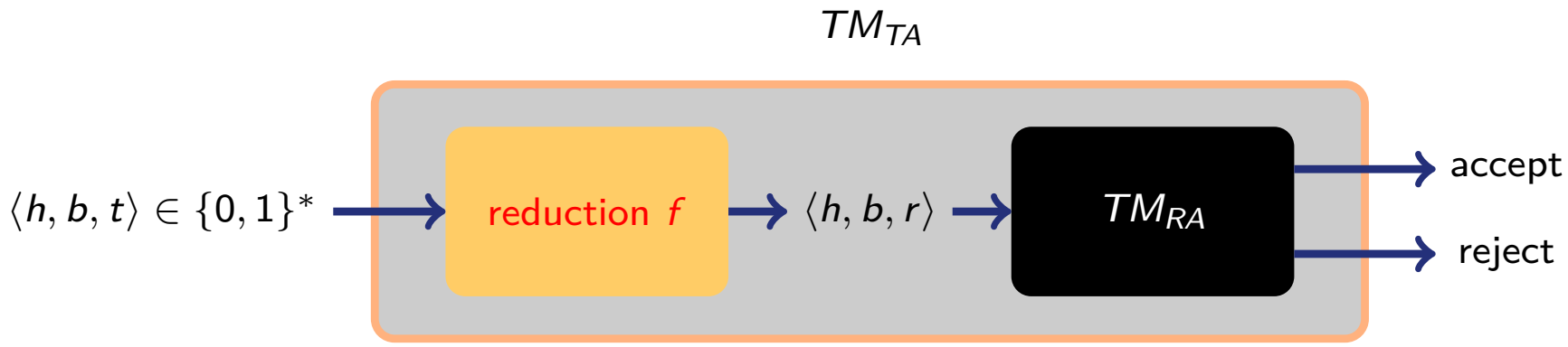

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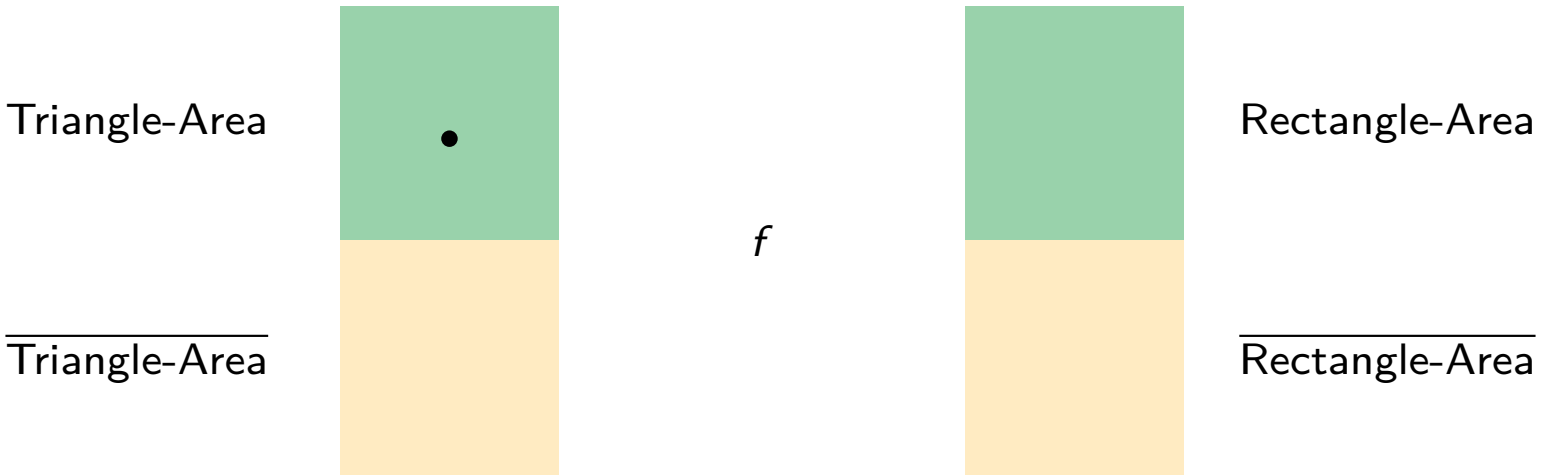

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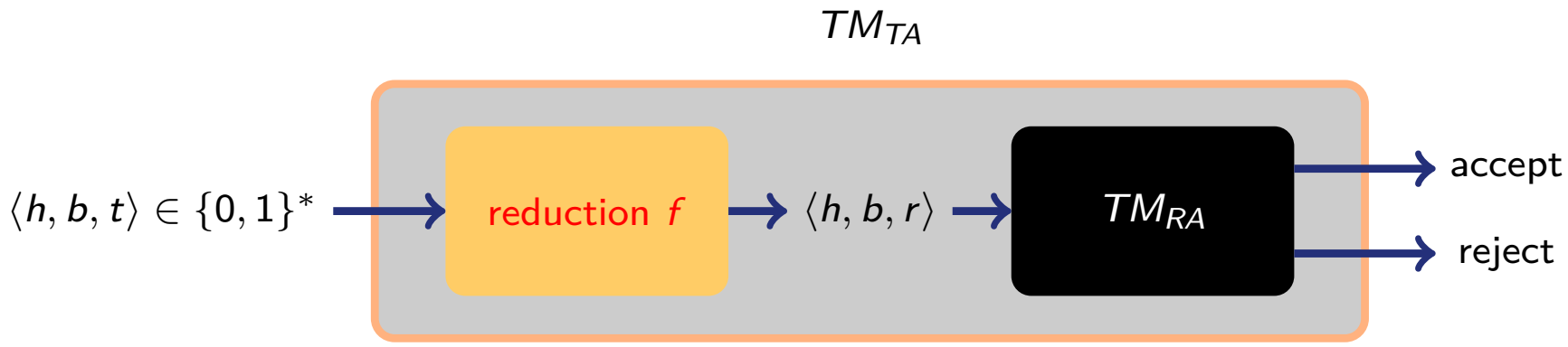

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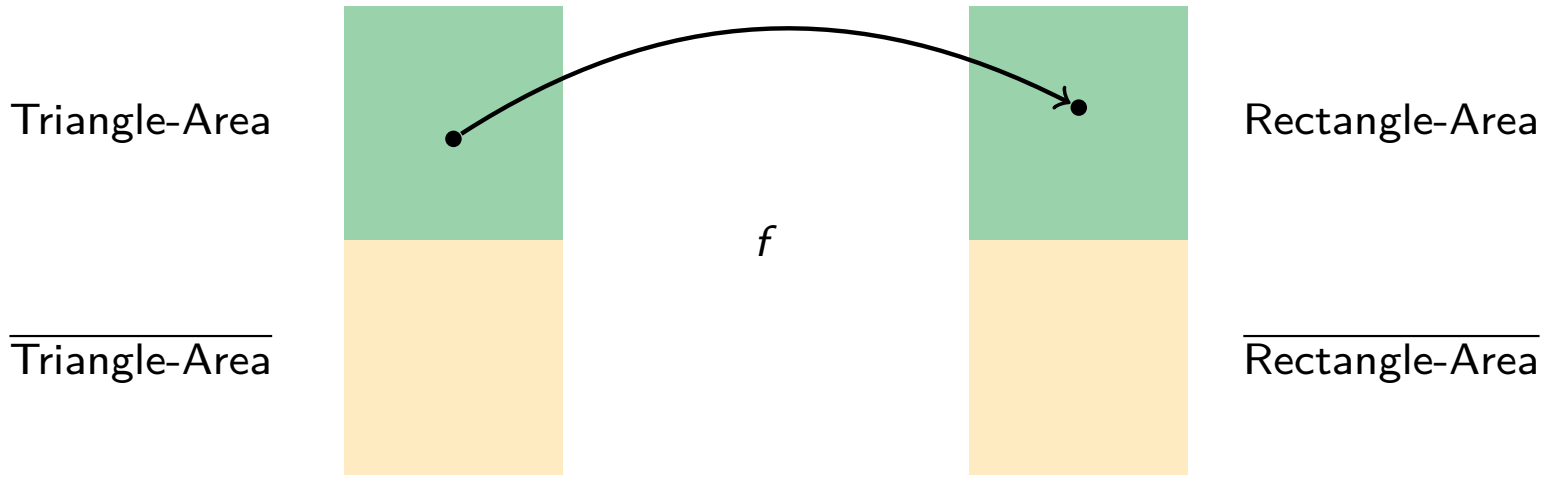

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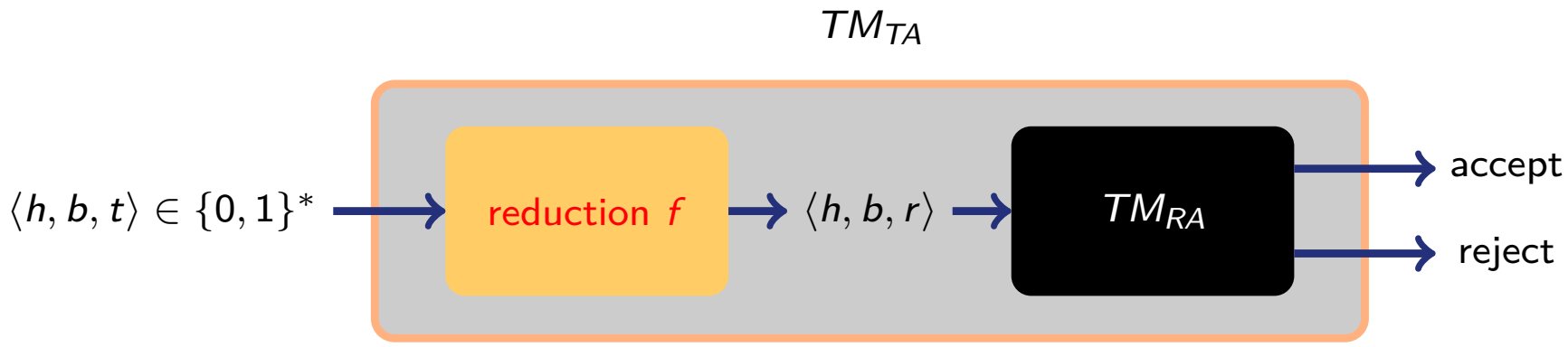

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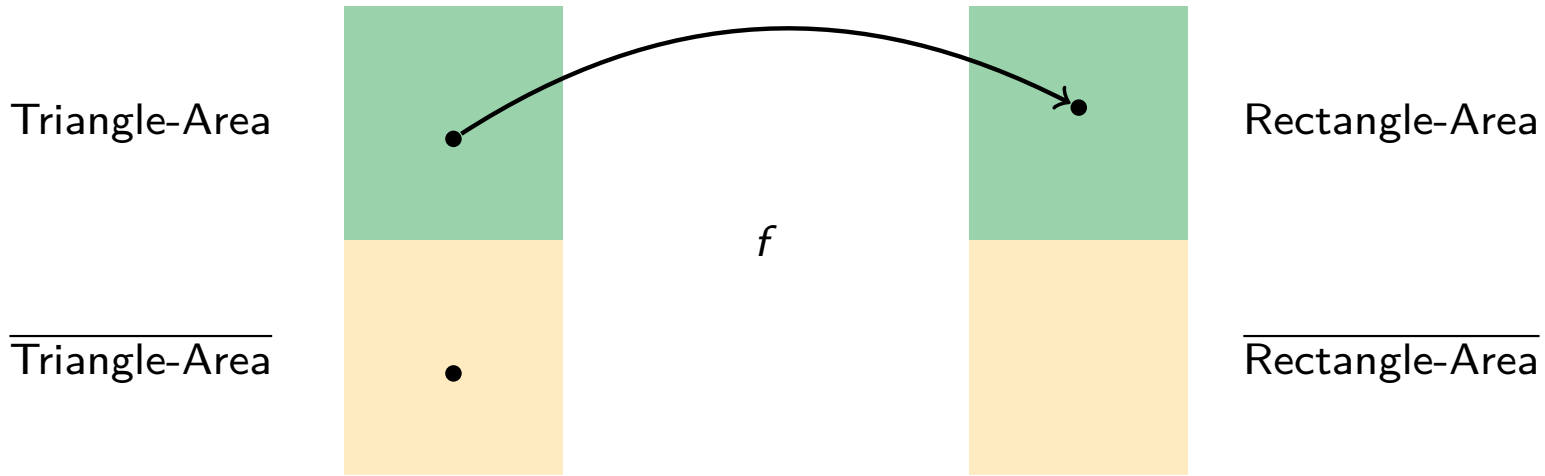

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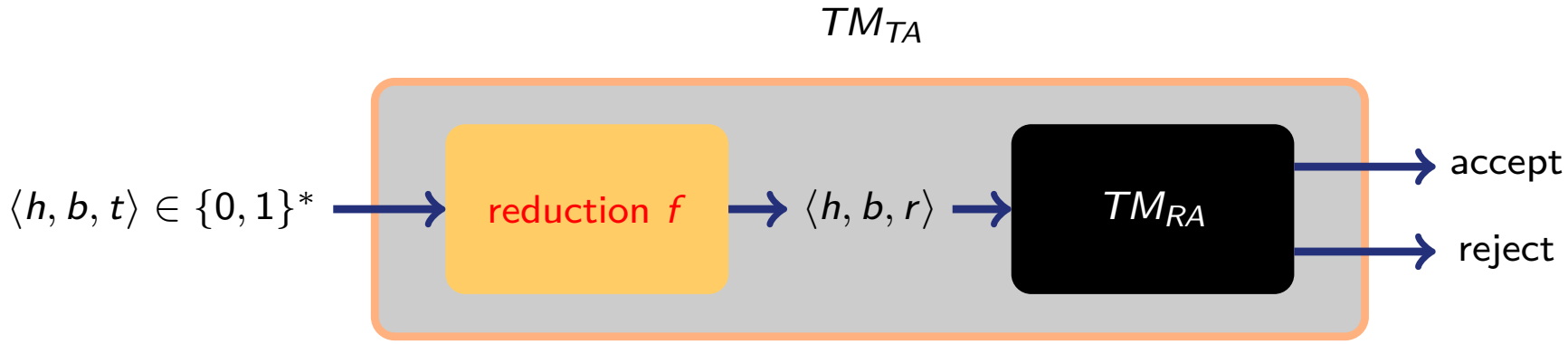

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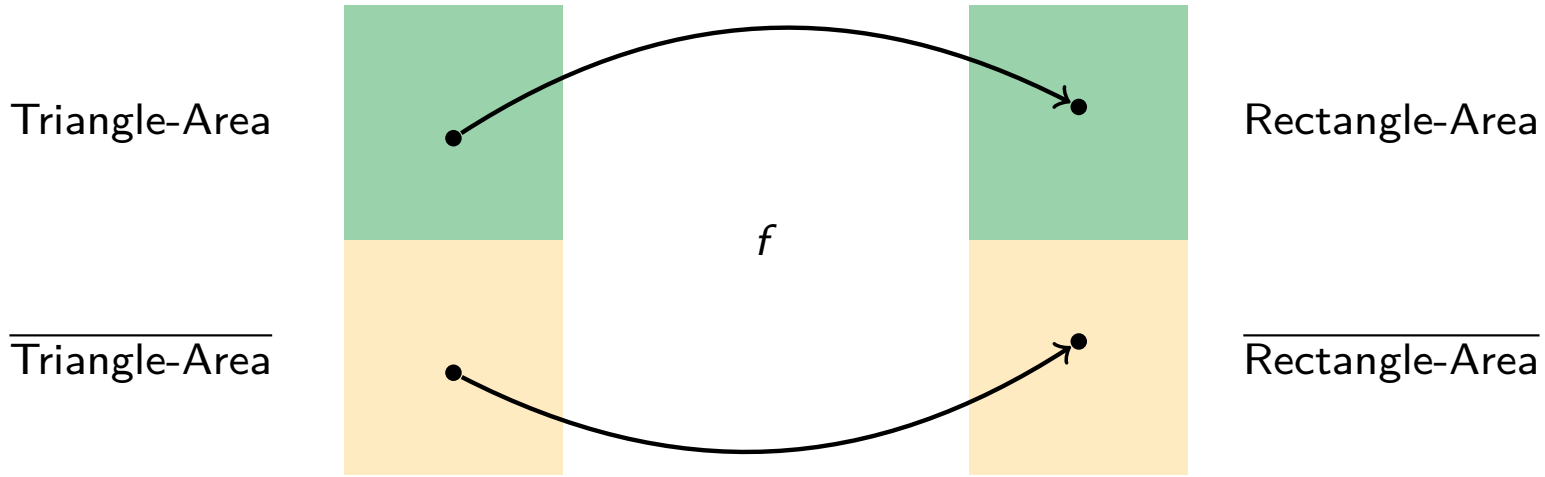

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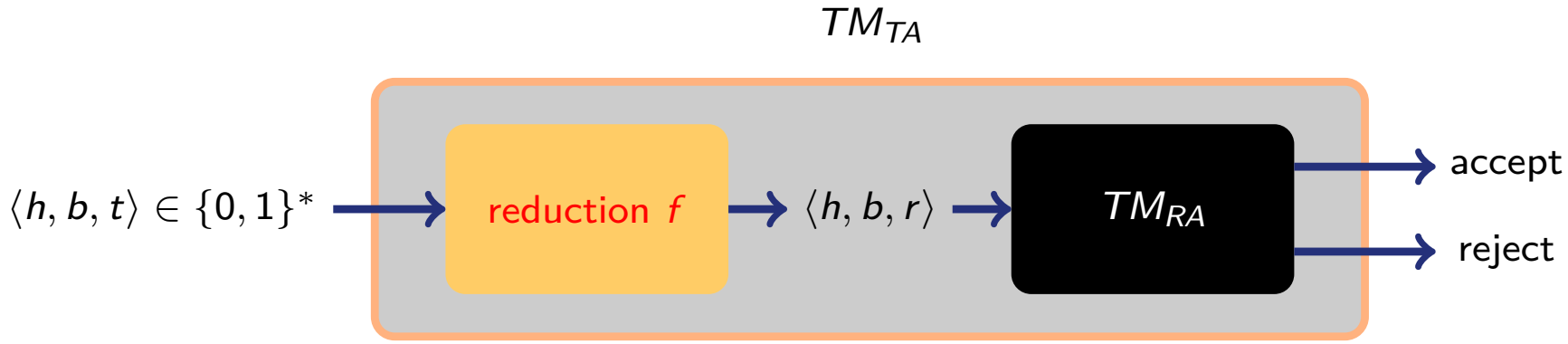

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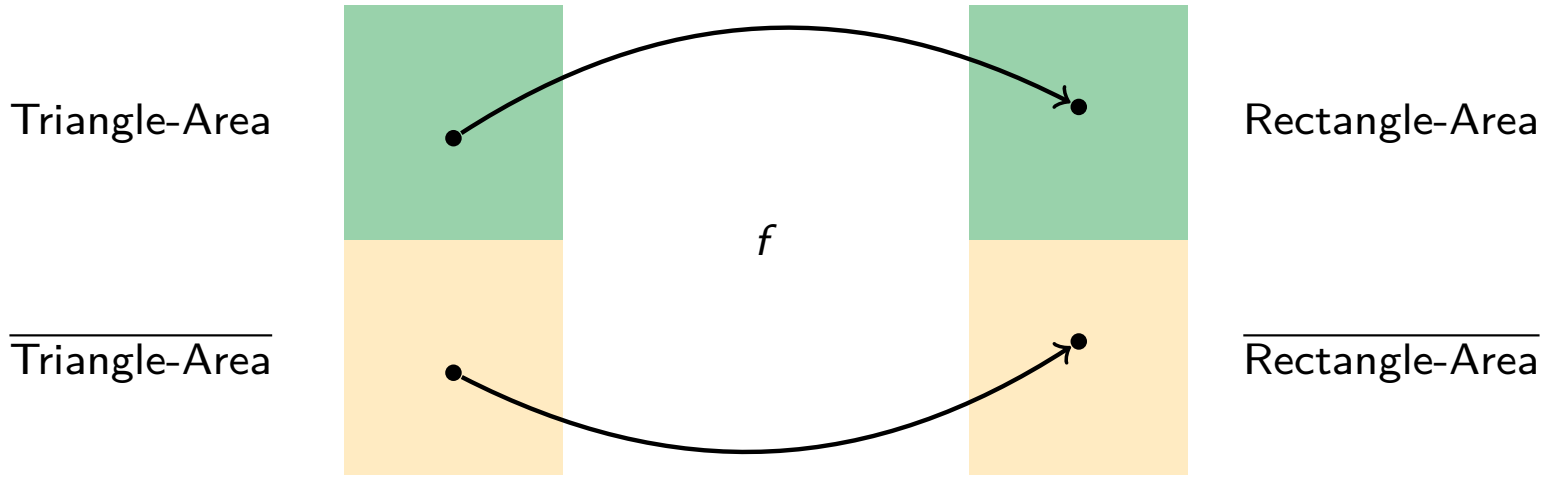

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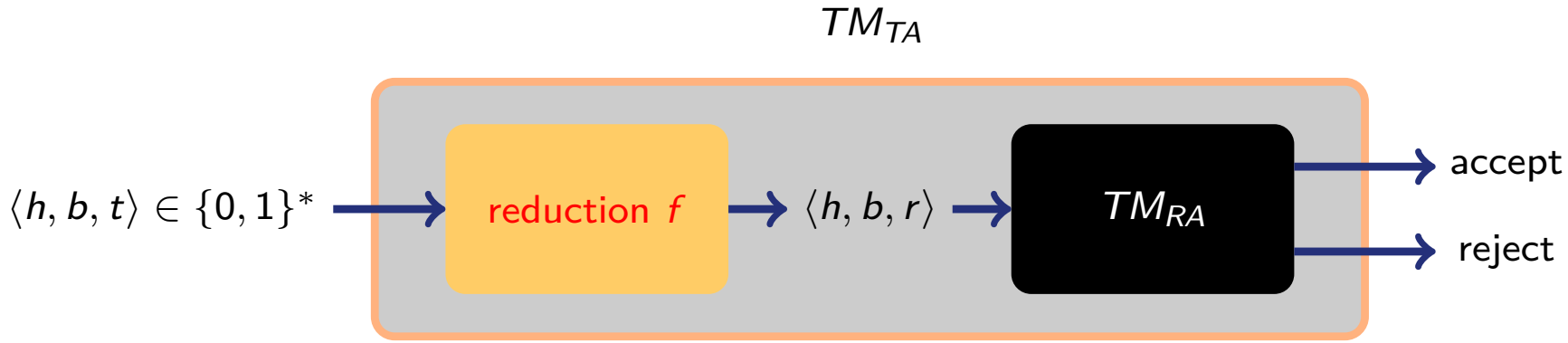

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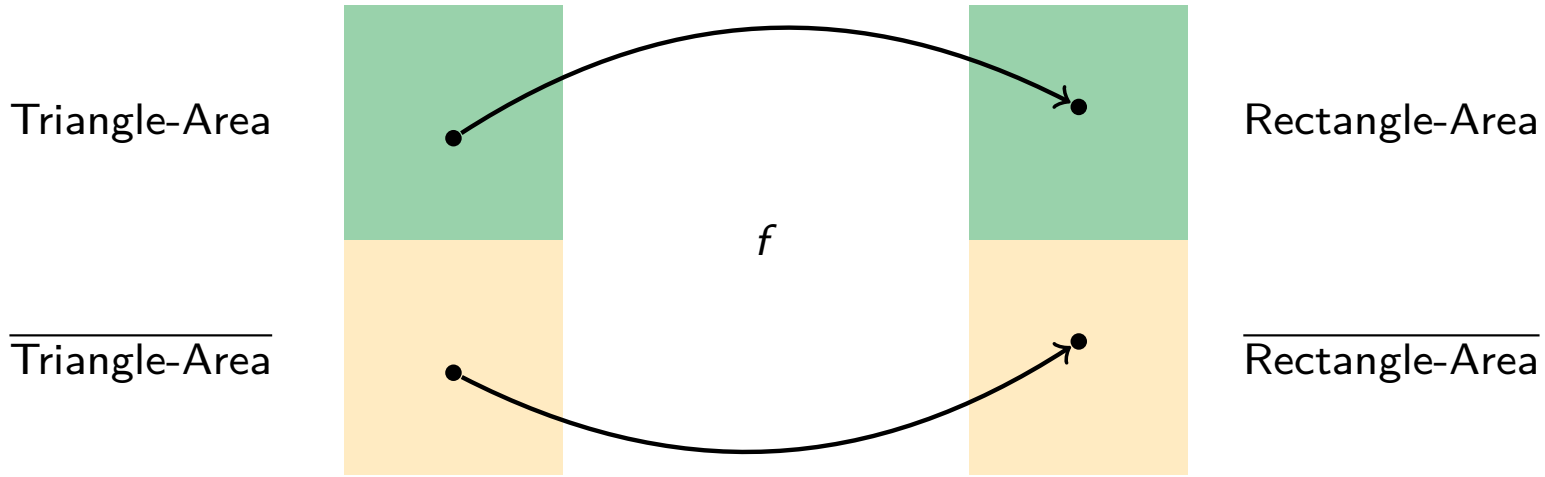
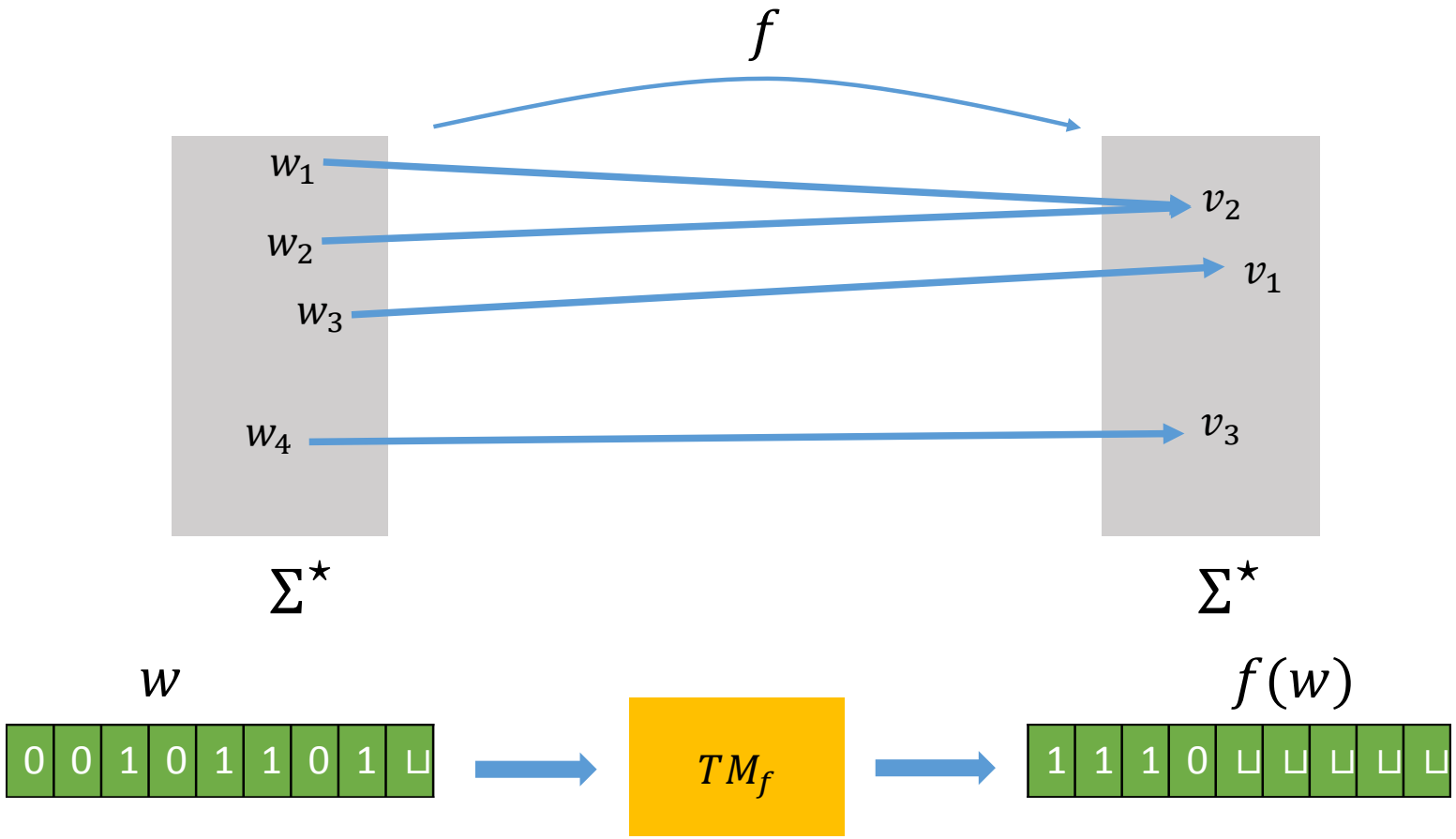


- f does not use the knowledge whether input is in language or not
- f can be computed with a TM

Mapping Reductions

Reductions Part 1: Computability



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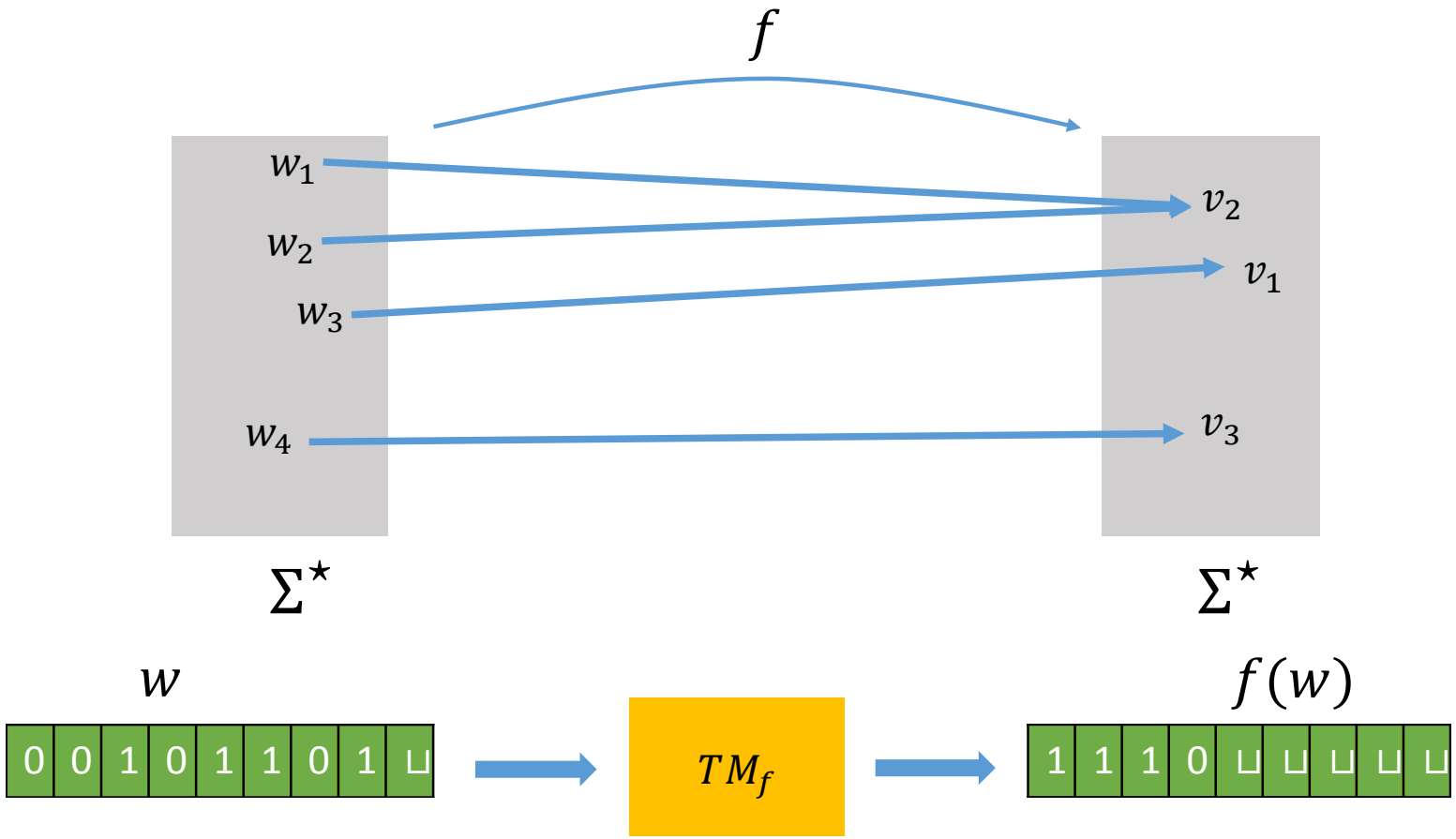
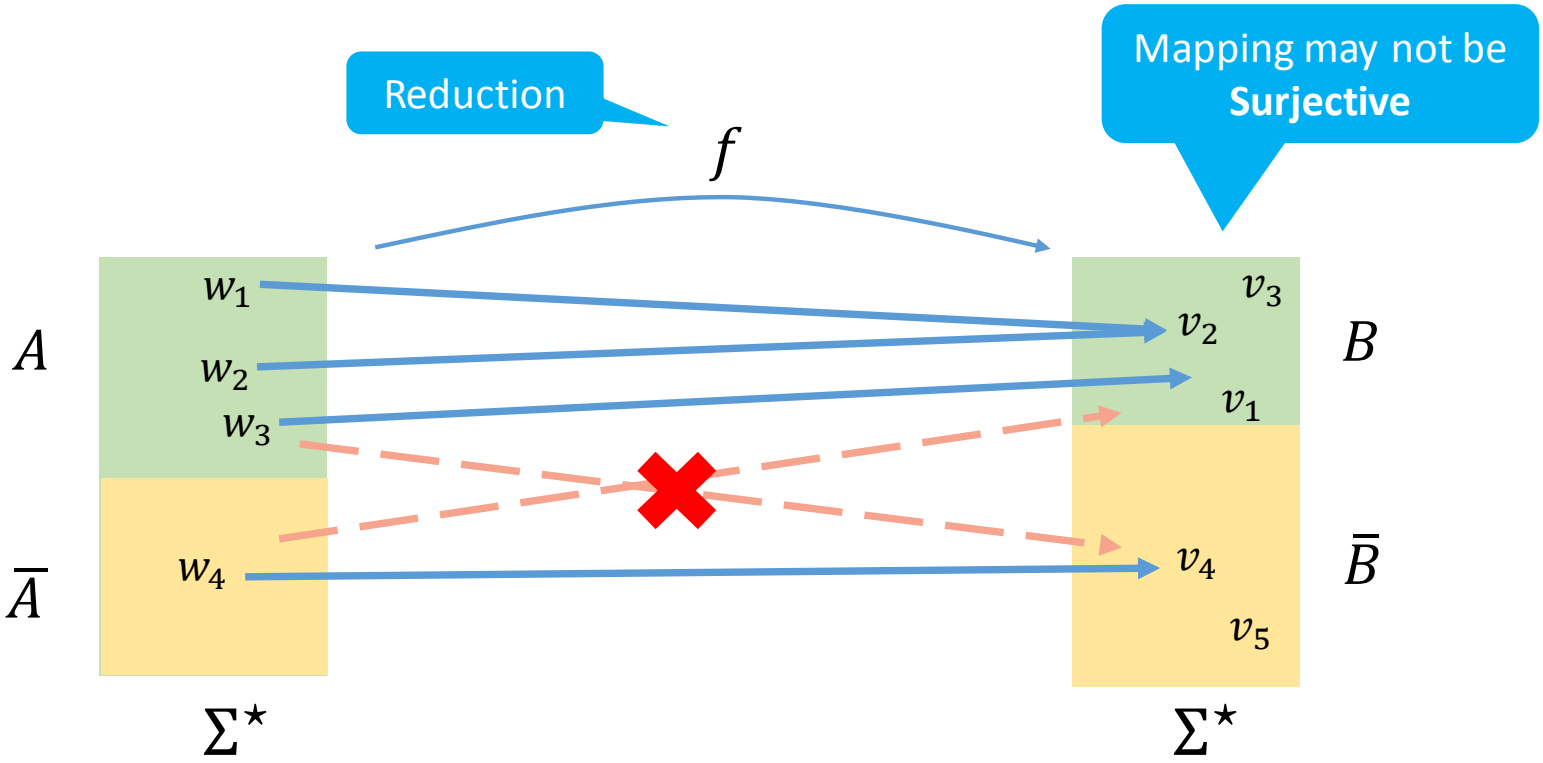


Definition: A function $f : \Sigma^* \to \Sigma^*$ is a *computable function* if some TM M, on every input w halts with just $f(w)$ on its tape

Reductions Part 2: Correctness



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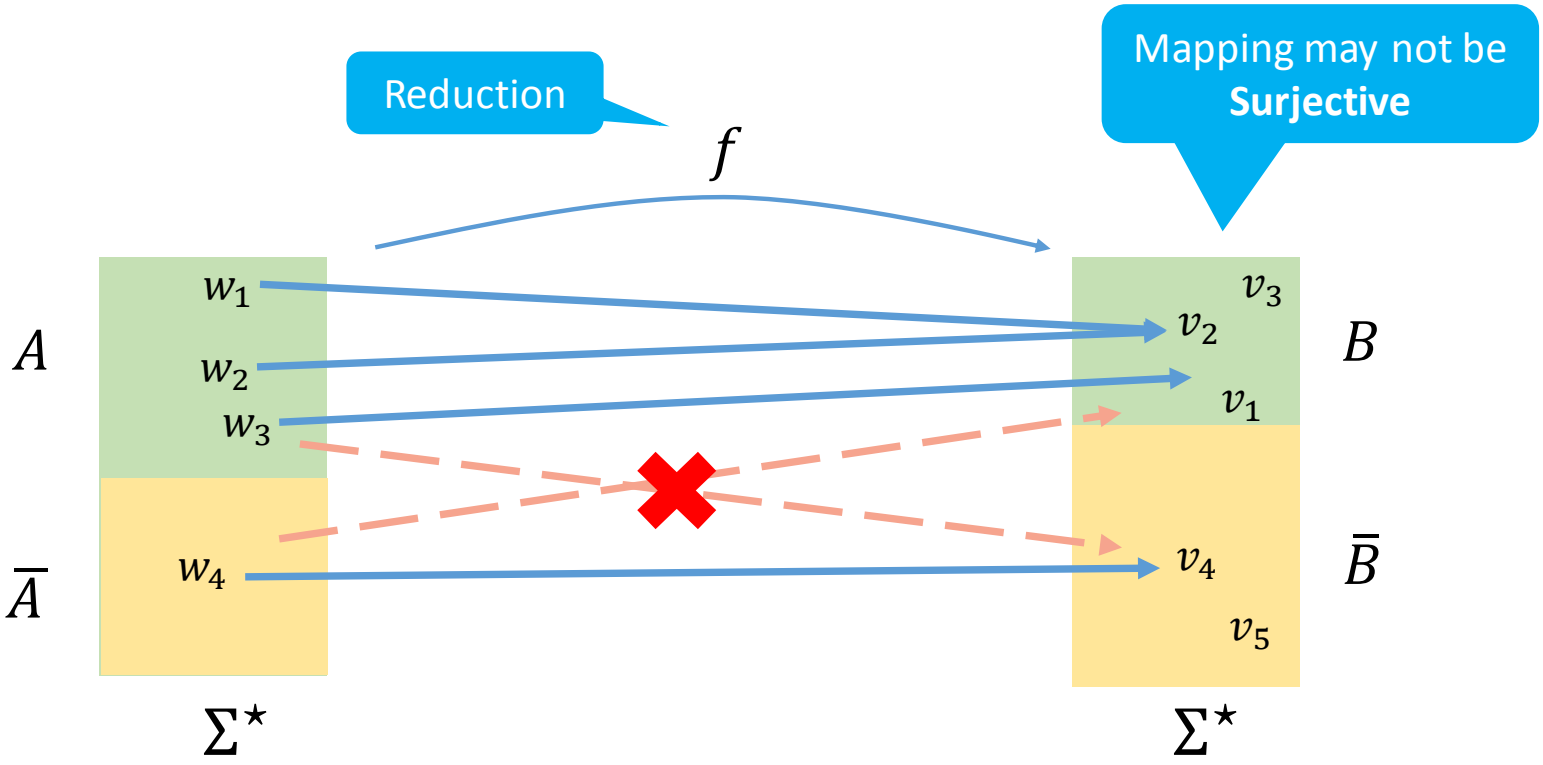

Definition: Language A is **mapping reducible** to language B, written $A \leq_m B$, if there is a computable function $f : \Sigma^* \to \Sigma^*$ such that for every $w \in \Sigma^*$:

$$w \in A \Leftrightarrow f(w) \in B$$

Theorem: If $A \leq_m B$ and B is decidable, then A is decidable

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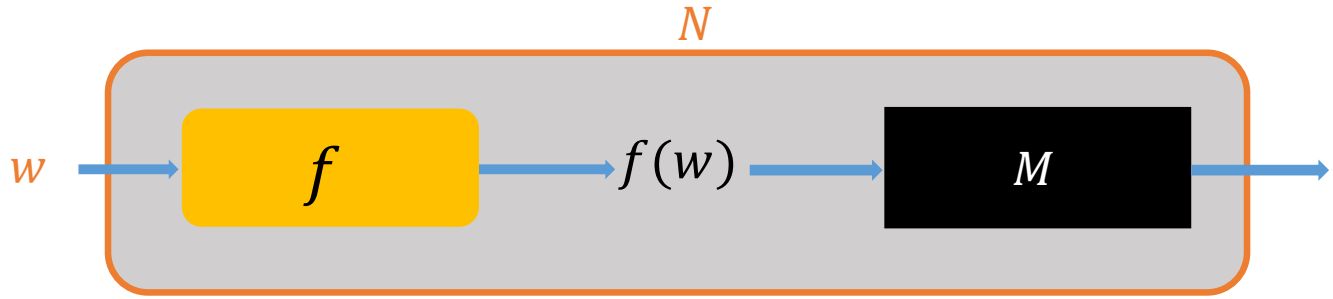

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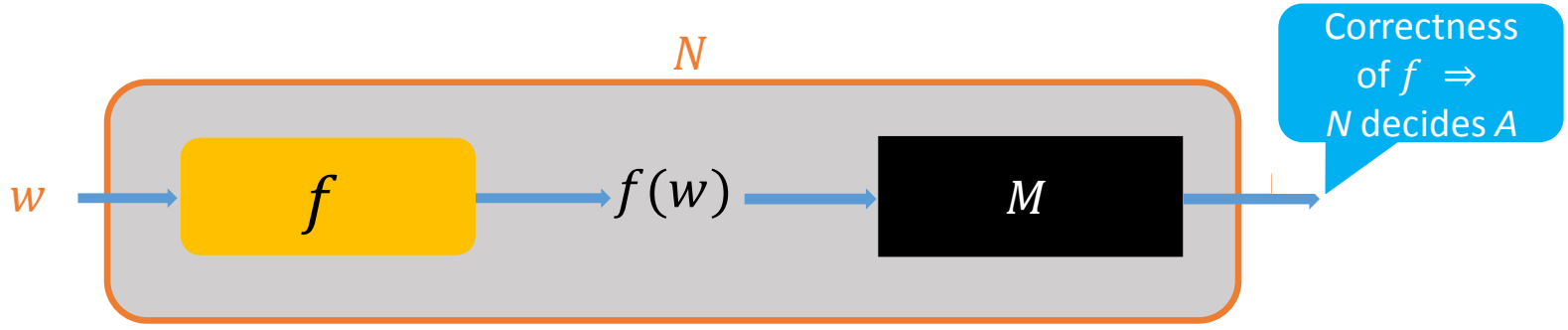

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Corollary: If $A \leq_m B$ and A is undecidable, then B is undecidable

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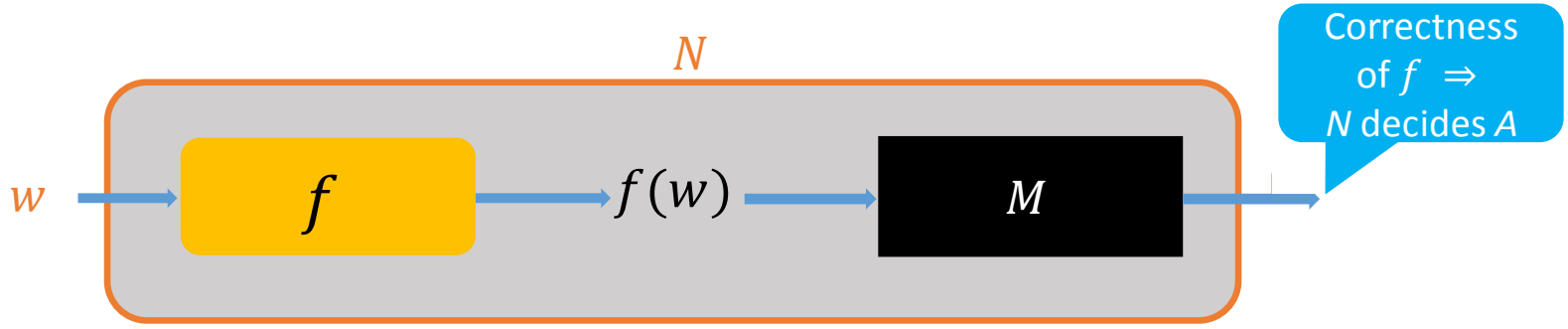

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Computability of f ⇒ N is a decider

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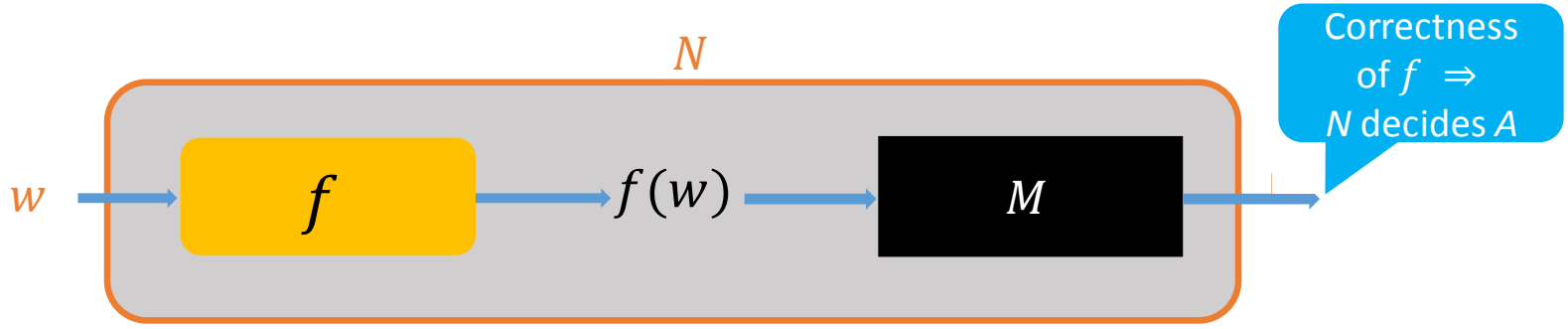

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Corollary: If $A \leq_m B$ and A is undecidable, then B is undecidable

Examples of Reductions

$$HALT_{TM} = \{\langle M, w\rangle \mid M \text{ halts on } w\}$$
$$A_{TM} = \{\langle M, w\rangle \mid M \text{ accepts } w\}$$

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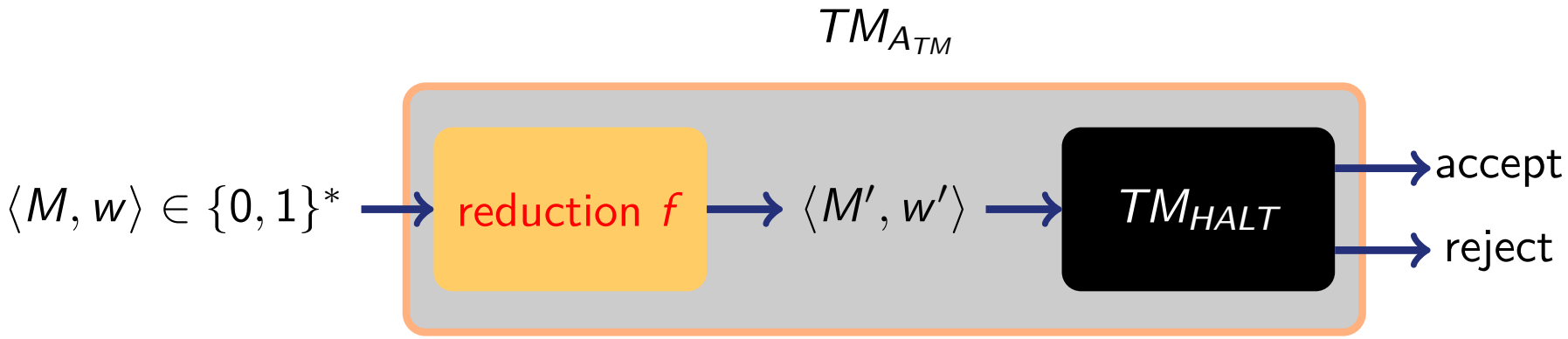

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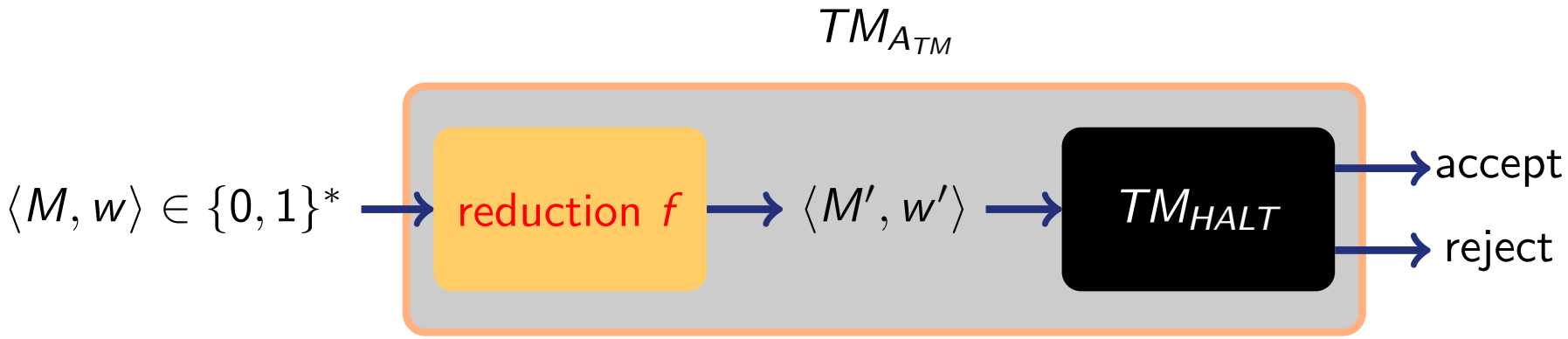
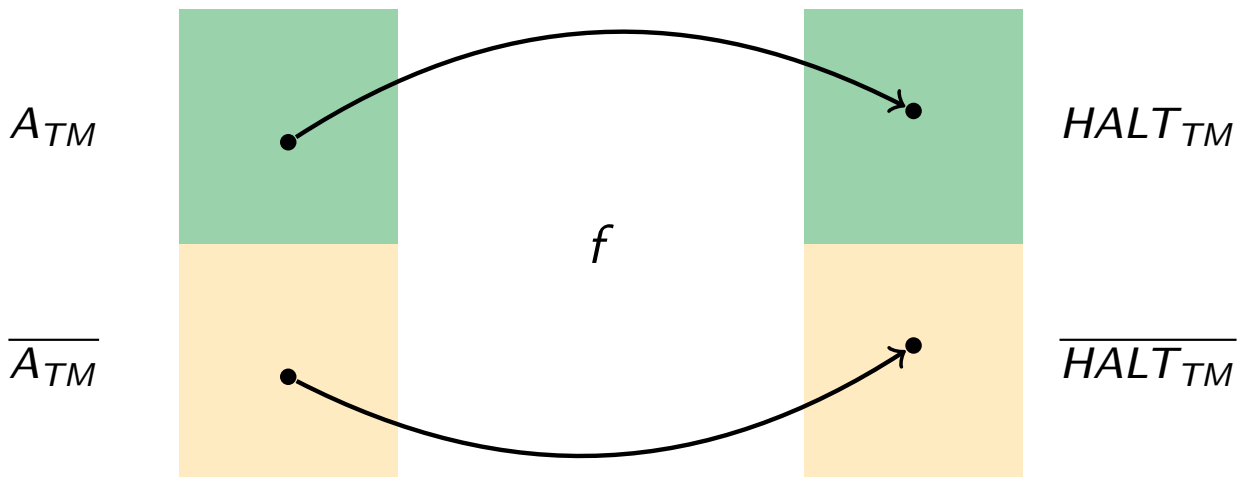


Define computable f such that



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- M' ="On input y :
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- $\Rightarrow$ If $\langle M, w \rangle \in A_{TM}$ then M' halts on w.
  Thus, $\langle M', w \rangle \in HALT_{TM}$
- $\Leftarrow$ If $\langle M, w \rangle \notin A_{TM}$ then either (1) will never halt or M rejects w and (2) will ensure no halting. Thus, $\langle M', w \rangle \notin HALT_{TM}$

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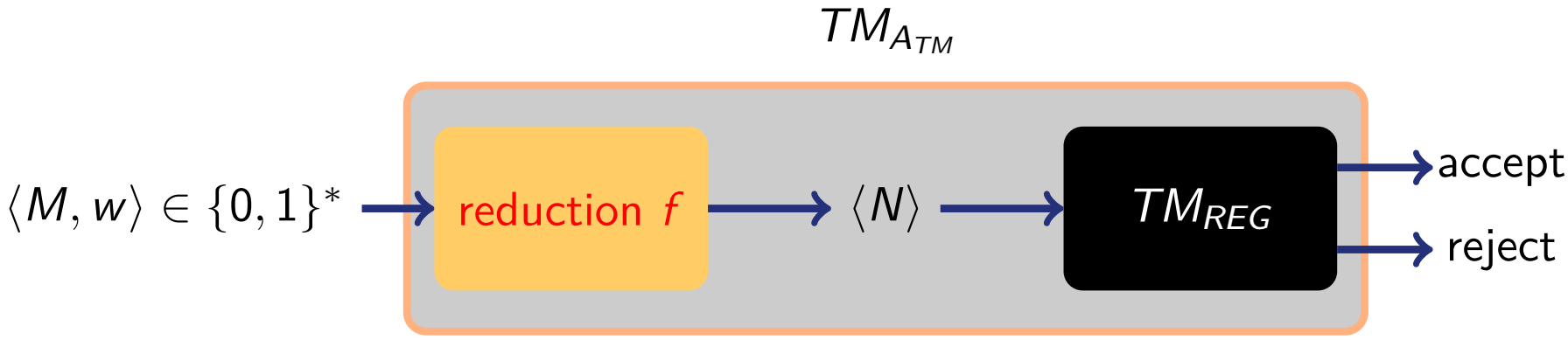

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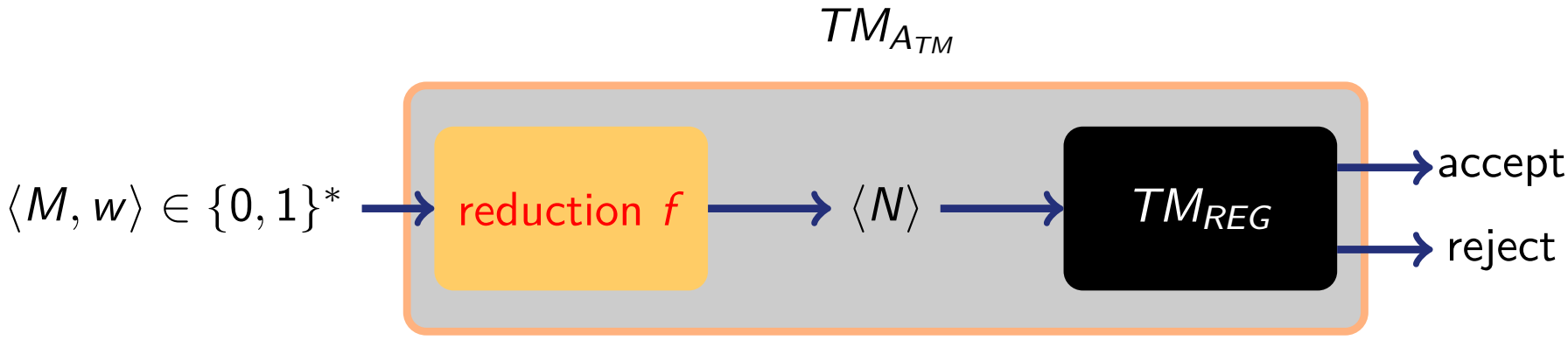

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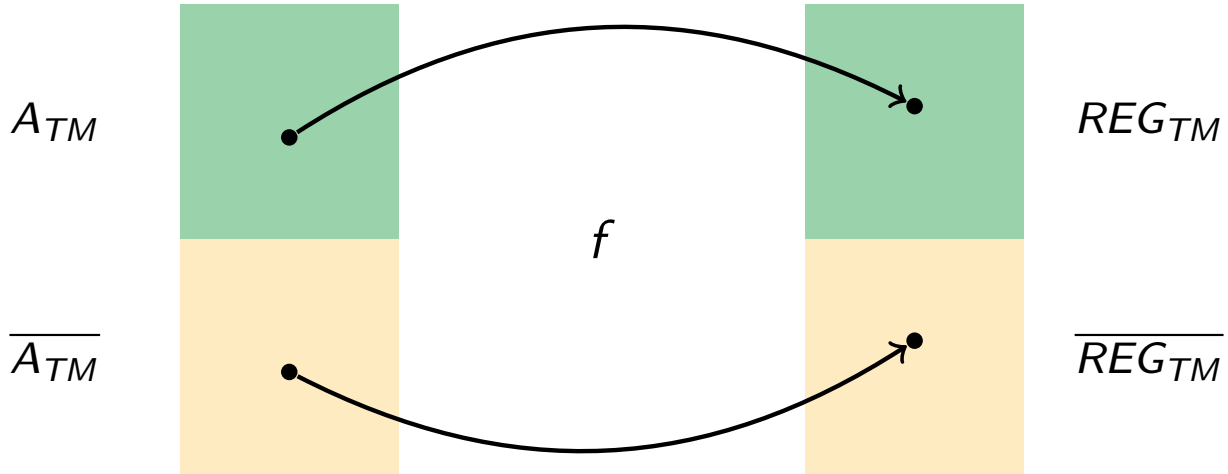

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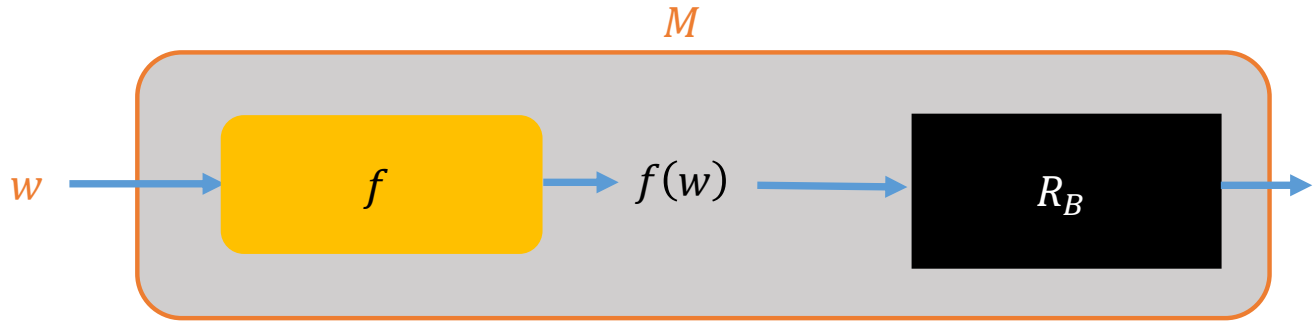
Theorem: If $A \leq_m B$ and B is recognizable then A is recognizable



Definition of M

R_B is a recognizer for B

$A \leq_m B$

M accepts w $\Leftrightarrow$ R_B accepts $f(w)$ $\Leftrightarrow$ $f(w) \in B$ $\Leftrightarrow$ $w \in A$

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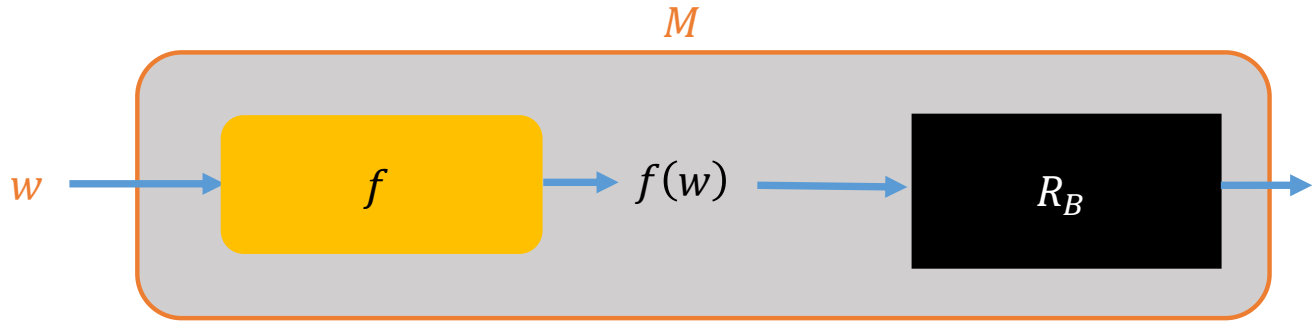

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Corollary: If $A \leq_m B$ and A is unrecognizable then B is unrecognizable

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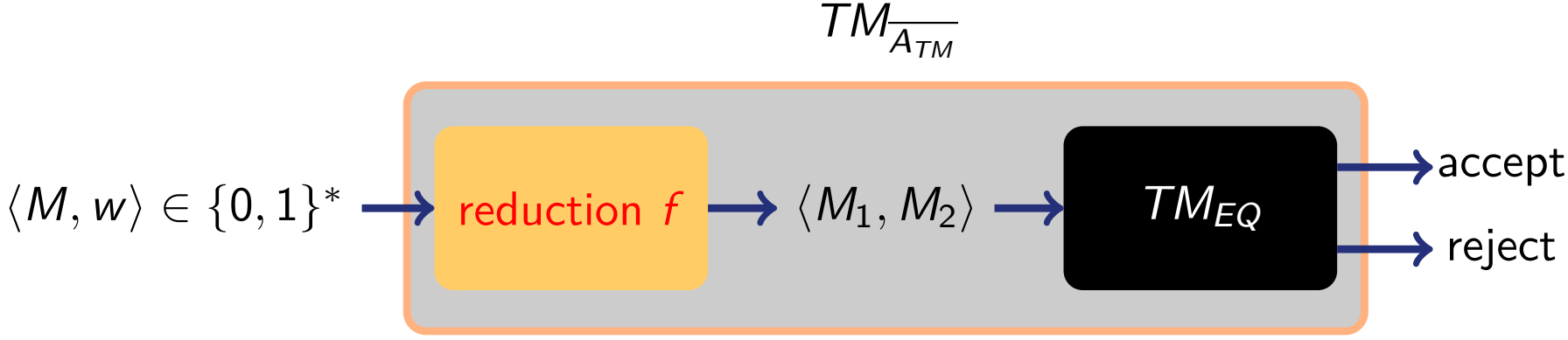

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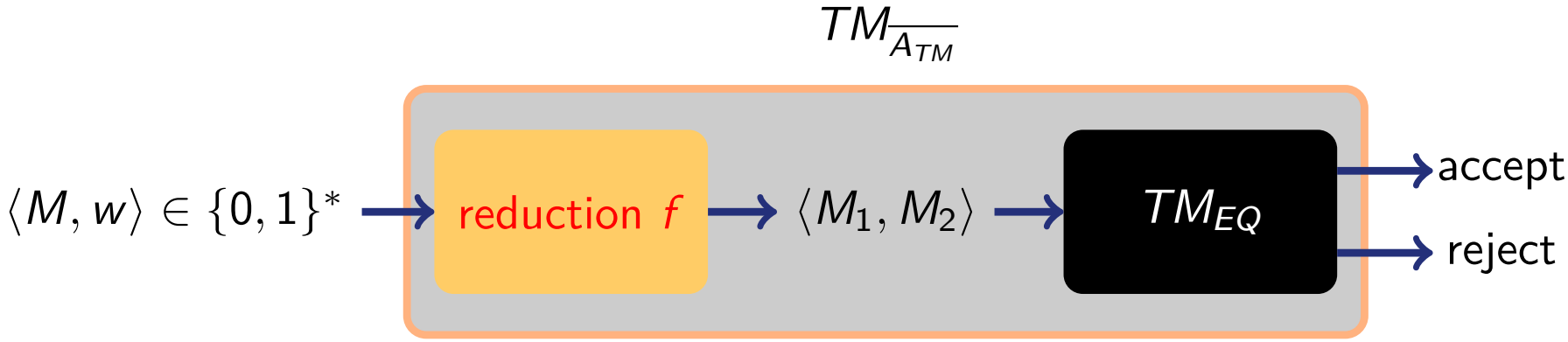

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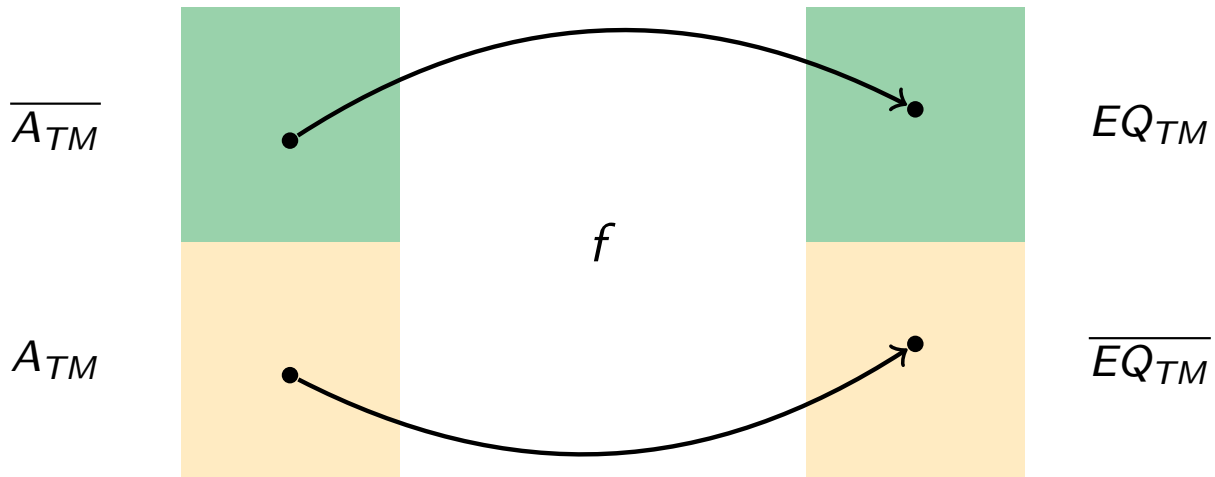

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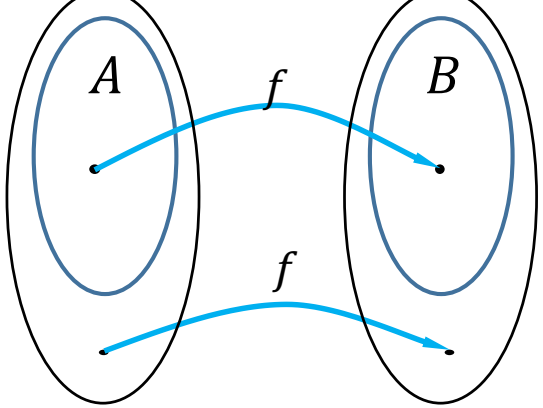
- $\Leftarrow$ If $\langle M, w\rangle \notin \overline{A_{TM}}$ then M accepts w and hence M_1 accepts every string. Thus $L(M_1) = \Sigma^* \neq \emptyset = L(M_2)$ — $\langle M_1, M_2\rangle \notin EQ_{TM}$

Summary of Reductions



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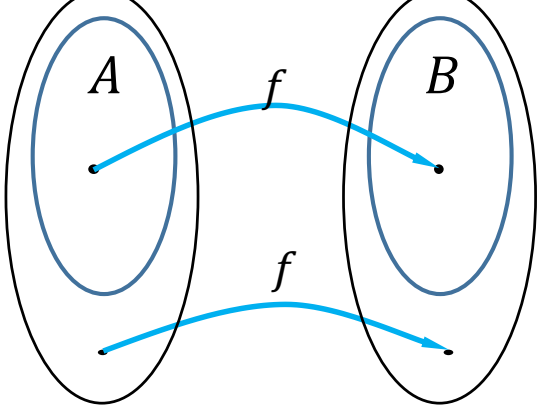

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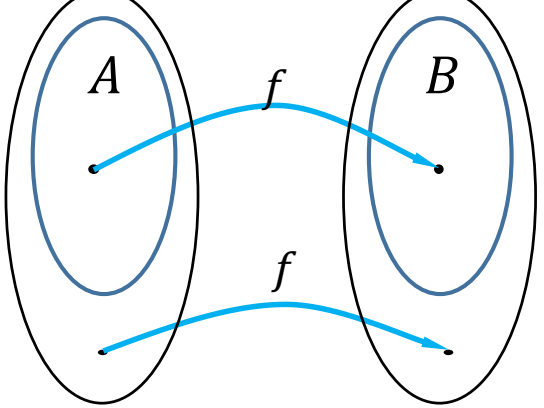

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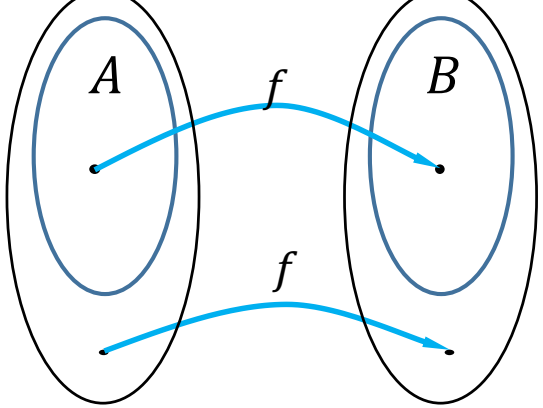

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Corollary: If $A \leq_m B$ and A is undecidable (unrecognizable) then B is undecidable (unrecognizable)

Some Exercises

True/False Questions

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True or false: A is decidable and $A \leq_m B \Rightarrow B$ is decidable

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- A is decidable so this function is computable

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- Then $x \in A \Leftrightarrow f(x) \in B$, so $A \leq_m B$.

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Take $A :=$ "any decidable nontrivial language" and $B := HALT_{TM}$.

True or false: This language is recognizable:

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  ($HALT_{TM}$ is recognizable and if $\overline{HALT_{TM}} \leq_m HALT_{TM}$ then $\overline{HALT_{TM}}$ would be recognizable, a contradiction)

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The same argument works if we start with $\overline{B}$ recognizable instead.

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$\overline{HALT_{TM}}$ is unrecognizable and $\overline{HALT_{TM}} \leq_m L^*$, so L^* is unrecognizable.
In particular, L^* is undecidable.

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Consider the languages

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- Suppose $\langle M, w \rangle \notin NR$; so M rejects w after some number t of steps. For every input $i \geq t$, step (2) makes M' loop forever. Hence M' does not halt on every input, so $\langle M' \rangle \notin AH$.